



3D PRINTING Emerging As GAME-CHANGER In Many Industries

Printing during Zero-G test flight by Space Foundry (Courtesy: Space Foundry)

Even as designers' creativity comes alive with 3D-printed clothes and jewellery at a fashion show, elsewhere in the world a cancer-ridden bird stays alive with a 3D-printed bill, and a new-born child gets a new lease on life with a 3D-printed skull. Indeed, the emerging applications of 3D printing are awe-inspiring and packed with hope



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Crescent, a 25-year-old Great Indian Hornbill, is one of the most loved birds in ZooTampa at Lowry Park, Florida (USA). When the zoo officials posted a picture of this winged beauty on their Instagram page last year with the tag line, "Crescent, our Great Indian Hornbill never misses a chance to strike a pose. Come out and visit her in our main aviary...", little would they have thought that she would be inflicted by a potentially fatal form of skin cancer called squamous cell carcinoma.

While the tumour is typically found near the front of the casque in hornbills, hers was in the back. This almost ruled out the possibility of a traditional extraction, as it would have exposed the bird's sinuses. So, a team at the University of South Florida's Radiology department printed a new casque using a Formlabs 3D printer and their new BioMed White Resin, a photocurable material that can be used to print biocompatible parts for prosthetic devices.

The material had just been developed by Formlabs, and they donated it to the

team trying to help Crescent. The team not only printed a new casque but also a replica of Crescent's head so that the team of veterinary surgeons could try the operation virtually before performing it on the bird—as it was a rare procedure, which had been done only once before by a veterinary surgeon in Singapore.

In January this year, the team removed the tumour and fitted the new casque on Crescent, who is now back in the aviary—as perky as ever. Luckily, it turned out that the resin was a perfect match for the yellow preening oils that keep a hornbill's casque bright and shiny, so she also looks as pretty as ever!

As against the early days of 3D printing, or additive manufacturing, when objects could be created only with a few polymer based materials, we now have advanced processes that enable 3D printing with dozens of materials ranging from carbon fibre composites, metals and alloys to sand, tungsten filament, and bio-compatible materials. Understandably, the applications and possibilities have also expanded manifold.

Industry 4.0 is gearing up to use additive manufacturing capabilities to print tools and